ARNAV DHARIYA

arnavdhariya24@gmail.com | (650) 684-4796 | in/arnavdhariya | https://arnavdhariya.github.io/

EDUCATION

University of California, Irvine June 2027

Bachelor of Science in Computer Science | Specialization in Intelligent Systems

Relevant Courses – Machine Learning (CS 178), AI (CS 171), Data Structures & Algorithms (ICS 46, CS 161), C/C++ (ICS 45C), Python (ICS 33)

Cumulative GPA: 3.92 | Dean's Honor List – **4** Quarters

EXPERIENCE

Sharad's Lab @ UCI August 2025 - Present

Undergraduate Researcher

- Designed and implemented differential privacy–based and ILP deletion mechanisms in Python with MySQL-backed experimental pipelines.
- Evaluated *Right-to-Be-Forgotten* guarantees on diverse UCI ML datasets through controlled empirical studies.
- Developed reproducible benchmarking framework for deletion efficiency and leakage analysis.
- Co-author on VLDB submission, in collaboration with PhD candidate Vishal Chakraborty.

The Zhang Lab at UCI August 2025 – Present

Undergraduate Researcher

- Investigated gene relationship datasets used in LLM- and ML-based biological models (rBio, PerturbQA).
- Designed evaluation pipelines using custom graph-based metrics to assess model accuracy across cell-line datasets.
- Developed a soft-verifier framework enabling weakly supervised training to reduce reliance on proprietary biological data.
- Contributed to interdisciplinary research team funded by UCI UROP ($2,000 award).

Network, Systems, and AI Lab (NetSAIL) August 2025 – Present

Undergraduate Researcher

- Built discrete-event simulation framework modeling large-scale sharding strategies (FAISS IVF, BigANN, BatANN).

- Evaluated system resilience under bursty workloads and server failure scenarios.
- Designed experiments to analyze load imbalance, failure propagation, and graceful degradation policies in distributed cloud infrastructure.
- Collaborated with PhD candidate Kapil Agrawal on scalable infrastructure research.

Computer Vision Lab August 2025 – January 2026

Undergraduate Researcher

- Conducted quantitative image analysis of hyperspectral and multispectral sensor data for environmental resilience applications.
- Contributed to Calit2-funded interdisciplinary project ($2,000 IRT award).
- Designed reproducible ML-based evaluation workflows for experimental imaging pipelines.

AngmarTek LLC June 2025 – August 2025

Grant Writer Intern

- Led drafting of NSF Phase I SBIR proposals in collaboration with UW Seattle, UIUC, DARPA-aligned programs, and federal agencies.
- Contributed to proposal strategy, technical sections, and commercialization narratives.
- Co-author on two in-progress publications.

INsite Lab @ UCI June 2025 – December 2025

Research Team Lead

- Directed 4-member graduate team (sole undergraduate lead) analyzing ICS departmental student data across academic and health metrics.
- Conducted stakeholder interviews to identify systemic gaps in mentorship and support structures.
- Presented data-driven recommendations to ICS Deans to inform policy-level improvement

Dr.Wayne Hayes Research Group September 2024 – September 2025

Research Team Lead

- Led simulated annealing–based haplotype assembly project as freshman researcher.
- Developed local optimization strategies (simulated annealing, hill climbing) to reconstruct haplotypes from fragmented DNA sequences.

- Implemented mechanisms to mitigate local minima and instability in alignment solutions.
- Presented research at UCI Undergraduate Research Symposium; recipient of the Chancellor's Award for Excellence in Undergraduate Research.
- https://uciurop.infoready4.com/#learnMore/28986

Green IT Lab

Researcher January 2025 – March 2025

Literature Reviewer

- Analyzed 300+ climate econometrics research papers, evaluating methodological rigor, reproducibility, and data/code availability.
- Synthesized findings into systematic review published on SSRN.

EXTRACURRICULARS

Indian Classical Music (Harmonium)

Competitive Recreational Cricket Player

Medium & Substack Technical Author

AWARDS

Undergraduate Research Symposium Finalist

Chancellor's Award for Excellence in Undergraduate Research

SKILLS

Languages: Python, C, C++, Java,

Tools/Technologies: Sci-kit Learn, PyTorch, MySQL, LaTex, Gurobi, Pulp, NumPy, Pandas

Key Strengths: Algorithm Design, Distributed Systems, Privacy-Preserving ML, Optimization Modeling, Leadership

PUBLICATIONS

[1] Freedman, Hayden and Foletta, Madison and Fernandez, Gilberto and Sokolova, Viktoriia and Dhariya, Arnav Abhijit and Hoek, André van der and Tomlinson, Bill, A Systematic Literature Review of Climate Econometrics Research (April 17, 2025). Available at SSRN: https://ssrn.com/abstract=6178246